

# Chemistry Simulator

## Complete Reference Guide

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Features: Elements | Compounds | Reactions | Thermal Analysis

### Document Overview

This PDF contains a comprehensive reference for the Chemistry Simulator including:

- â ¢ All chemical elements with atomic properties
- â ¢ Common compounds and their formulas
- â ¢ Reaction data with thermal properties
- â ¢ Enthalpy changes (H) and temperature changes (T)
- â ¢ Classification of exothermic and endothermic reactions

Total Elements: 20

Total Compounds: 12

Total Reactions: 10

# Chemical Elements

## Metals

Fe - Iron | Atomic #: 26 | Mass: 55.85  
Cu - Copper | Atomic #: 29 | Mass: 63.55  
Zn - Zinc | Atomic #: 30 | Mass: 65.39  
Al - Aluminum | Atomic #: 13 | Mass: 26.98  
Na - Sodium | Atomic #: 11 | Mass: 22.99  
K - Potassium | Atomic #: 19 | Mass: 39.1  
Ca - Calcium | Atomic #: 20 | Mass: 40.08  
Mg - Magnesium | Atomic #: 12 | Mass: 24.31  
Au - Gold | Atomic #: 79 | Mass: 196.97  
Ag - Silver | Atomic #: 47 | Mass: 107.87  
Pt - Platinum | Atomic #: 78 | Mass: 195.08

## Non-Metals

H - Hydrogen | Atomic #: 1 | Mass: 1.008  
C - Carbon | Atomic #: 6 | Mass: 12.01  
N - Nitrogen | Atomic #: 7 | Mass: 14.01  
O - Oxygen | Atomic #: 8 | Mass: 16.0  
S - Sulfur | Atomic #: 16 | Mass: 32.07  
P - Phosphorus | Atomic #: 15 | Mass: 30.97  
Cl - Chlorine | Atomic #: 17 | Mass: 35.45  
Br - Bromine | Atomic #: 35 | Mass: 79.9  
I - Iodine | Atomic #: 53 | Mass: 126.9

# Chemical Compounds

H<sub>2</sub>O (Water) [OXIDE]  
CO<sub>2</sub> (Carbon Dioxide) [OXIDE]  
NaCl (Sodium Chloride) [SALT]  
HCl (Hydrochloric Acid) [ACID]  
H<sub>2</sub>SO<sub>4</sub> (Sulfuric Acid) [ACID]  
HNO<sub>3</sub> (Nitric Acid) [ACID]  
NaOH (Sodium Hydroxide) [BASE]  
KOH (Potassium Hydroxide) [BASE]  
NH<sub>3</sub> (Ammonia) [BASE]  
CuSO<sub>4</sub> (Copper Sulfate) [SALT]  
FeCl<sub>3</sub> (Iron(III) Chloride) [SALT]  
AgNO<sub>3</sub> (Silver Nitrate) [SALT]

# Chemical Reactions - Thermal Data

## High Energy Reactions (Exothermic)

Reaction:  $\text{K} + \text{H}_2\text{O}$

Equation:  $2\text{K} + 2\text{H}_2\text{O} \rightarrow 2\text{KOH} + \text{H}_2$

$H = -393 \text{ kJ/mol}$  |  $T = \sim 150^\circ\text{C}$

Reaction:  $\text{Na} + \text{H}_2\text{O}$

Equation:  $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$

$H = -368 \text{ kJ/mol}$  |  $T = \sim 130^\circ\text{C}$

Reaction:  $\text{Mg} + \text{O}_2$

Equation:  $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$

$H = -1204 \text{ kJ/mol}$  |  $T = \sim 200^\circ\text{C}$

Reaction:  $\text{H}_2 + \text{O}_2$

Equation:  $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$

$H = -285 \text{ kJ/mol}$  |  $T = \sim 280^\circ\text{C}$

Reaction:  $\text{C} + \text{O}_2$

Equation:  $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$

$H = -393 \text{ kJ/mol}$  |  $T = \sim 85^\circ\text{C}$

## Medium Energy Reactions

Reaction:  $\text{Mg} + \text{HCl}$

Equation:  $\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2$

$H = -99 \text{ kJ/mol}$  |  $T = \sim 35^\circ\text{C}$

Reaction:  $\text{Zn} + \text{HCl}$

Equation:  $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$

$H = -102 \text{ kJ/mol}$  |  $T = \sim 38^\circ\text{C}$

Reaction:  $\text{NaHCO}_3 + \text{CH}_3\text{COOH}$

Equation:  $\text{NaHCO}_3 + \text{CH}_3\text{COOH} \rightarrow \text{NaCH}_3\text{COO} + \text{H}_2\text{O} + \text{CO}_2$

$H = -56 \text{ kJ/mol}$  |  $T = \sim 12^\circ\text{C}$

## Other Reactions

Reaction: Fe + O<sub>2</sub>

Equation:  $4\text{Fe} + 3\text{O}_2 \rightarrow 2\text{Fe}_2\text{O}_3$

H = -1648 kJ/mol | T = ~850C

Reaction: NaOH + HCl

Equation:  $\text{NaOH} + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O}$

H = -57 kJ/mol | T = ~25C

# Thermal Reference

## Key Thermal Concepts

Exothermic Reaction: Releases heat to surroundings ( $H < 0$ )  
Endothermic Reaction: Absorbs heat from surroundings ( $H > 0$ )  
Enthalpy Change (H): Heat energy change per mole  
Temperature Change (T): Predicted temperature increase/decrease

## Energy Classifications

High Energy:  $T > 100^{\circ}\text{C}$  | Violent reactions (explosions, intense fires)  
Medium Energy:  $40^{\circ}\text{C} < T \leq 100^{\circ}\text{C}$  | Vigorous reactions with significant heat  
Low Energy:  $0^{\circ}\text{C} < T \leq 40^{\circ}\text{C}$  | Mild reactions with gentle heat release  
No Energy:  $T = 0^{\circ}\text{C}$  | No reaction or color-change only

## Chemistry Simulator Features

Real-time thermal simulation with accurate calculations  
Interactive reaction visualization  
Laboratory apparatus simulation  
Educational calorimetry experiments

## Statistics

Total Chemical Substances: 32  
Combinatorial space:  $>10^{20}$   
Exothermic Reactions Documented: 8/10